A mini course in network meta-analysis

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Session contents

- Recap information on meta-analysis
- Theory of NMA
- Minimal example of NMA
- More complicated models of (N)MA
- Discussion of use cases of NMA

Basic terminology (recap)

- effect size,
- generalisability,
- heterogeneity,
- fixed effects,
- random effects,
- meta-regression,
- forest plot

recap: previous slides + hierarchical model interpretation of MA + meta-analytic weights

- pooling factor

What if we want to compare many treatments?

Our motivating example: Cipriani *et al.*, *Comparative efficacy and acceptability of antimanic drugs in acute mania: a multiple-treatments meta-analysis* (2011), the Lancet

- Influential paper (600+ citations)
- Meta-analysis experts
- Realistic scope
- Highlights typical outputs
- . . . but also challenges

Network meta-analysis

- Loosely: any meta-analysis with more than two comparators
- Direct comparisons, many treatments (studies with 2+ arms)
- Indirect comparisons: "traversing the graph"
- Also known as ITC, MTC
- MA: pairwise or direct
- NMA: Multi-arm studies, pairwise studies with more than 2 treatments

Acceptability of NMA

- "New kid on the block": much younger than MA
- See e.g. "Is network meta-analysis as valid as standard pairwise meta-analysis?" (2013)
- Most widely developed/promoted in health technology assessment, e.g. NICE DSU

Simple case: A, B, C

Let's draw a simple example.

Additional assumptions

- Joint randomisability ("you'd put them in the same study")
- Transitivity (same populations/effect modifiers)
- Consistency ("take any path in the graph")

Transitivity is hard to test for. Consistency a bit easier, but low power.

If not assuming FE:

- Heterogeneity (same or different across each pairwise comparison?)
- Within trial correlations

NMA outputs

Revisit Cipriani et al. paper

- Pairwise comparison matrix
- Forest plots
- Rankograms

Implementation

- MA has simple Bayesian and frequentist implementations
- Situation a bit more complicated than in the case of MA.
- Frequentist in R: `netmeta` (good, mature software)
- But many regulators (rightly) prefer Bayesian methods
- Bayesian in R: `gemtc` (mature, but slightly less user-friendly)

Some situations require bespoke models.

More complicated situations

- Reminder: (in multi-arm studies) we typically meta-analyse SMD, OR, RR, HR
- But you can write *any* parameteric model in Bayesian inference
- . . . and analyse it hierarchically
- Consider a model of survival or a PKPD model
- . . . just need data to infer these parameters
- Individual level data often inferred from summary-level data
- Blurry line between meta-analysis and hierarchical modeling

In pharma world this is referred by some as model-based meta-analysis

Examples of survival analyses

- Example 1: nivolumab vs cabozantinib
- Example 2: cabozantinib vs SoC

Worked example in WinBUGS (link):

```
Model{
    for (i in 1:N){ # N=number of data points in dataset
        #likelihood
        r[i]~ dbin(p[i],n[i])
        p[i]<-1-exp(-h[i]*dt[i])

        log(h[i])<-nu[i] + log(time[i])*theta[i]
        nu[i]<-mu[s[i]]+delta[s[i]]*(1-equals(t[i],b[i]))
        theta[i]<-0
        }

    for(k in 1 :NS){ # NS=number of studies in dataset
        delta[k] <- md[k]
        md[k]<-d[ts[k]]-d[bs[k]]
    }

    # priors
        d[1]<-0
    for(j in 2 :NT){ # NT=number of treatments
        d[j] ~ dnorm(mean[1],prec2[1,1])
    }
    for(k in 1 :NS){
        mu[k] ~ dnorm(mean[1],prec2[1,1])
    }
}
```

A few resources that recap NMA for you

- Nice online book by Matthais Harrier:
  - Intro to frequentist NMA
  - Intro to Bayesian NMA (gemtc)
- Big textbook: Dias et al. (2018)
- Cipriani et al. (2011)
- NICE DSU technical support document on Bayesian NMA: directly relevant to the regulator, oldie but goodie
  - Also see subsequent documents (vol 3-4; this is vol. 2)
- NMA of parametric survival curves (Ouwens paper)
  - survivalnma package